

Technology #112

Apply topology optimization (SIMP) to
shoulder prosthesis components to minimize
weight while preserving stiffness and
structural safety without active control

Engineering Technology-Transfer Dossier

elements
CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Apply topology optimization (SIMP) to shoulder prosthesis components to minimize weight while preserving stiffness and structural safety without active control elements

WHY DOES IT MATTER?

Documented recurring problem 'appropriate_device_problem_term_code_not' for shoulder prosthesis: 23 MAUDE reports across 3 distinct devices (20200506 to 20211104); root-cause evidence from 2 recall records and 10 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 2 recall root-cause descriptions (FDA enforcement records, custody attached); 10 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, cust Unresolved gap: Across the 3 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: appropriate, available, conditions, devices, events, failure, function, injury, intervention, introducing, maintain, matching, modes, must, producing, recurrin

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 12 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in mechanical_structural. 10-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 10-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Fatigue testing of topology-optimized components versus standard components under cyclic loading simulating 10 years of use. Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: buckling / kink margin tests across duty cycle. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design; superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead; K for catheter constraint conditions is MODEL_DERIVED; MEC-002 (Section bending stress) is applicable only under: linear-elastic material; small deflection; prismatic section.; MEC-003 (Hooke's law (linear elasticity)) is applicable only under: below proportional limit; isothermal.; MEC-001 (Euler buckling load) is applicable only under: elastic, slender column, small deflection.; Model input variables requiring sourced values before any numeric result: M (N.m), c (m), I (m⁴), E (Pa), epsilon (-), E (Pa), I (m⁴), K (-), L (m) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: sigma (Pa), sigma (Pa), P_cr (N) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): member slenderness ratio [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; yield/ultimate strength [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; applied cyclic load amplitude [P_cr, N] value_status=UNKNOWN; required run-out cycles [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; articulation radius [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; friction coefficient at guides [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Passive structural optimization with topology optimization

POTENTIAL VALUE DRIVER: Addresses: Documented recurring problem 'appropriate_device_problem_term_code_not' for shoulder prosthesis: 23 MAUDE reports across 3 distinct devices (20200506 to 20211104); root-cause evidence from 2 recall records and 10 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 2 recall root-cause descriptions (FDA enforcement records, custody attached); 10 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, cust Unresolved gap: Across the 3 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: appropriate, available, conditions, devices, events, failure, function, injury, intervention, introducing, maintain, matching, modes, must, producing, recurrin

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: mechanical_structural

Disciplines: mechanical engineering, structural analysis, fatigue and fracture, materials science

Architecture: PROPOSED architecture (no physical system exists): load path / structural backbone; actuation or compliance element; guidance / bearing interfaces; fatigue-critical features register — realizing: Apply topology optimization (SIMP) to shoulder prosthesis components to minimize weight while preserving stiffness and structural safety without active control elements

Subsystems:

?: load path / structural backbone - ?

?: actuation or compliance element - ?

?: guidance / bearing interfaces - ?

?: fatigue-critical features register - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Apply topology optimization (SIMP) to shoulder prosthesis components to minimize weight while preserving stiffness and structural safety without active control elements

Key Physics: Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

MEC-002: Section bending stress. Expression: $\sigma = M * c / I$. Applicability to this invention: CONDITIONAL; holds under: linear-elastic material; small deflection; prismatic section. Assumptions: stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design. Engaged variables: NOT ESTABLISHED.

MEC-003: Hooke's law (linear elasticity). Expression: $\sigma = E * \epsilon$. Applicability to this invention: CONDITIONAL; holds under: below proportional limit; isothermal. Assumptions: superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead. Engaged variables: NOT ESTABLISHED.

MEC-001: Euler buckling load. Expression: $P_{cr} = \pi^2 * E * I / (K * L)^2$. Applicability to this invention: CONDITIONAL; holds under: elastic, slender column, small deflection. Assumptions: K for catheter constraint conditions is MODEL_DERIVED. Engaged variables: NOT ESTABLISHED.

Assumptions:

- stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design
- superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead
- K for catheter constraint conditions is MODEL_DERIVED
- MEC-002 (Section bending stress) is applicable only under: linear-elastic material; small deflection; prismatic section.
- MEC-003 (Hooke's law (linear elasticity)) is applicable only under: below proportional limit; isothermal.
- MEC-001 (Euler buckling load) is applicable only under: elastic, slender column, small deflection.
- Model input variables requiring sourced values before any numeric result: M (N.m), c (m), I (m^4), E (Pa), epsilon (-), E (Pa), I (m^4), K (-), L (m) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: sigma (Pa), sigma (Pa), P_cr (N) — UNKNOWN until the inputs are sourced and the model is evaluated.

- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): member slenderness ratio [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; yield/ultimate strength [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; applied cyclic load amplitude [P_cr, N] value_status=UNKNOWN; required run-out cycles [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; articulation radius [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; friction coefficient at guides [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'appropriate device problem term code not', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'kink/buckling below required push ability', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'fatigue crack initiation at stress concentrator', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'wear debris generation at sliding interfaces', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'joint separation under overload', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'set-release of superelastic shape after set', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	Devices matching 'shoulder prosthesis' must maintain function under the conditions of a recurring injury events with	SDRRC-FACT
DI-002	Mechanism driver (proposed)	Passive structural optimization with topology optimization	MODELLED
DI-003	Expected effect target	Reduce mechanical failures through optimized load distribution while maintaining structural integrity under repetitive load	MODELLED
DI-004	Domain design input pattern: delivery path applicability and NOT ESTABLISHED (candidate pattern; confirm again)	ENGINEERING_PROPOSED	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: actuation applicability and NOT ESTABLISHED (candidate pattern; confirm again)	ENGINEERING_PROPOSED	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: duty cycle applicability and NOT ESTABLISHED (candidate pattern; confirm again)	ENGINEERING_PROPOSED	ENGINEERING_PROPOSED
DI-007	Regulatory design input: compliance candidate standards (ASTM F2516 (ratio design) pathway UNKNOWN (NEVER PROPOSED)	Dis never inferred	Dis never inferred
DI-008	Custodied observation: The LIBRARY Need List: Hydro-Pneumatic/38202932/medical-3839446e7f6208a2300519e10254827a76b185b45312ed		
DI-009	Custodied observation: An In vitro Study https://europepmc.org/article/med/32280669 Motions of wash 278ac769c5b542b20ef9a03afad141e814fcd1227f868		
DI-010	Custodied observation: Development of a Prosthetic Rehydrostatic/39771673 system to the shoulder 13e38044061e54035679f36e7fa43d7bf64e267		
DI-011	Custodied observation: Design and Test https://www.researchgate.net/publication/314714981 Control of a 6f8c989f1b050240323f3ad145b1628b5f53c4bc2c8d47		
DI-012	Custodied observation: Brain-Computer Interface https://www.researchgate.net/publication/344233669 is not trained 70d1a38e455348c556ba03bada7f265c747295026932		

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	load path / structural backbone realizing Apply topology optimization (SIMP) to shoulder prosthesis components to minimize weight while preserving stiffness	CONCEPTUAL	shoulder prosthesis components to minimize weight while preserving stiffness
DO-002	actuation or compliance element realizing Apply topology optimization (SIMP) to shoulder prosthesis components to minimize weight while preserving stiffness	CONCEPTUAL	shoulder prosthesis components to minimize weight while preserving stiffness
DO-003	guidance / bearing interfaces realizing Apply topology optimization (SIMP) to shoulder prosthesis components to minimize weight while preserving stiffness	CONCEPTUAL	shoulder prosthesis components to minimize weight while preserving stiffness
DO-004	fatigue-critical features register realizing Apply topology optimization (SIMP) to shoulder prosthesis components to minimize weight while preserving stiffness	CONCEPTUAL	shoulder prosthesis components to minimize weight while preserving stiffness
DO-005	load path / structural backbone interfaces with actuation or compliance element: signal energy amplification (dimensions, apply topology optimization (SIMP) to shoulder prosthesis components to minimize weight while preserving stiffness	CONCEPTUAL	signal energy amplification (dimensions, apply topology optimization (SIMP) to shoulder prosthesis components to minimize weight while preserving stiffness
DO-006	actuation or compliance element interfaces with guidance / bearing interfaces: signal energy amplification (dimensions, apply topology optimization (SIMP) to shoulder prosthesis components to minimize weight while preserving stiffness	CONCEPTUAL	signal energy amplification (dimensions, apply topology optimization (SIMP) to shoulder prosthesis components to minimize weight while preserving stiffness
DO-007	guidance / bearing interfaces interfaces with fatigue-critical features register: signal energy amplification (dimensions, apply topology optimization (SIMP) to shoulder prosthesis components to minimize weight while preserving stiffness	CONCEPTUAL	signal energy amplification (dimensions, apply topology optimization (SIMP) to shoulder prosthesis components to minimize weight while preserving stiffness
DO-008	Device boundaries: patient/fluid inlet and outlet terminal control/data points required by Apply topology optimization (SIMP) to shoulder prosthesis components to minimize weight while preserving stiffness	CONCEPTUAL	patient/fluid inlet and outlet terminal control/data points required by Apply topology optimization (SIMP) to shoulder prosthesis components to minimize weight while preserving stiffness
DO-009	Flow/force/signal path geometry realizing the mechanism is in Apply topology optimization (SIMP) to shoulder prosthesis components to minimize weight while preserving stiffness	CONCEPTUAL	mechanism is in Apply topology optimization (SIMP) to shoulder prosthesis components to minimize weight while preserving stiffness
DO-010	Operating range for member slenderness ratio within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-011	Operating range for yield/ultimate strength within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-012	Operating range for applied cyclic load amplitude within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-013	Operating range for required run-out cycles within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-014	Operating range for articulation radius within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-015	Operating range for friction coefficient at guides within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-016	Sensing -> decision -> actuation loop maintaining the mechanism operating position for Apply topology optimization (SIMP) to shoulder prosthesis components to minimize weight while preserving stiffness	PROPOSED	mechanism operating position for Apply topology optimization (SIMP) to shoulder prosthesis components to minimize weight while preserving stiffness
DO-017	Information path: sensing element -> decision/monitor control/act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical in	CONCEPTUAL	sensing element -> decision/monitor control/act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical in
DO-018	Fixture enabling: buckling / kink margin tests across duty cycle rig concept named, rigid specification UNKNOWN	UNKNOWN	buckling / kink margin tests across duty cycle rig concept named, rigid specification UNKNOWN
DO-019	Fixture enabling: fatigue life (run-out) testing — rig concept named, build specific instrumentation selection; acceptance instrumentation	UNKNOWN	fatigue life (run-out) testing — rig concept named, build specific instrumentation selection; acceptance instrumentation
DO-020	Fixture enabling: friction and wear characterization — rig concept named, build specific geometry, UNKNOWN	UNKNOWN	friction and wear characterization — rig concept named, build specific geometry, UNKNOWN

ID	Description	Status	Missing Inputs
DO-021	Fixture enabling: buckling/kink margin test across duty cycle	UNKNOWN	UNKNOWN

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
member slenderness ratio	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned and target) value and verification method k
yield/ultimate strength	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned and target) value and verification method k
applied cyclic load amplitude	UNKNOWN (no sourced value)	N	ENGINEERING_PROPOSED	(design assigned and target) value and verification method k
required run-out cycles	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned and target) value and verification method k
articulation radius	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned and target) value and verification method k
friction coefficient at guides	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned and target) value and verification method k

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
appropriate device provided by appropriate	Failure by appropriate device	designer controls the STAB system (requires designer to know how to use)	None; triggered on the operating order (Atrib)	
kink/buckling below rated load	Failure by kink/buckling below rated load	compressive strength, axial elongation, stress and support capacity (EN 606) has been checked	calculated fatigue load (F _{fat})	
fatigue crack initiation	Failure by fatigue crack initiation	material stress, free geometry, use of S-N curves for notches, laser peening, etc. not integrated	calculated failure fatigue (A _{dur})	
wear debris generation	Failure by wear debris generation	material, design, end use, size, feed rate, pressure, etc. not fully verified	calculated failure debris (A _{dur})	
joint separation under load	Failure by joint separation under load	external load, size, material, feed rate, etc. not fully verified	calculated failure debris (A _{dur})	
set-release of superlattice	Failure by set-release of superlattice	shape, set, process, etc. not fully verified	calculated failure debris (A _{dur})	

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
appropriate device problem	Fails by appropriate device problem	design control the DEVICES TABLE SHALL require the device work single activation the operating (And NOT)	
kink/buckling below required	Edgely kink/buckling below required	possibility of axial buckling under the design load. Specimen test the axial buckling mode. (And NOT)	
fatigue crack initiation at stress	Fails by fatigue crack initiation at stress	concentration of the stress at surface. Use the design stress for the fatigue life. (And NOT)	
wear debris generation at sliding	Sliding wear debris generation at sliding	ate the debris and debris removal. Use the design debris removal rate. (And NOT)	
joint separation under overload	Fails by joint separation under overload	ation of a larger sized cover load test the joint design. (And NOT)	
set-release of superelastic shape	Shape set-release of superelastic shape	after the stress period. Use the design stress period. (And NOT)	

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Fatigue testing of topology-optimized components using state-of-the-art optimization algorithms and design standards (e.g., ASME, SAE, etc.)	Finite element analysis (FEA) using advanced optimization algorithms (e.g., genetic algorithms, etc.)	Acceptance criteria: No failure observed under specified loading conditions.	Result: No failure observed under specified loading conditions.
VF-002	run killer experiment: falsification_test	run killer experiment (e.g., falsification_test)	run killer experiment (e.g., falsification_test)	run killer experiment (e.g., falsification_test)
VF-003	buckling / kink margin tests across duty cycle	buckling / kink margin tests (e.g., buckling / kink margin tests)	buckling / kink margin tests (e.g., buckling / kink margin tests)	buckling / kink margin tests (e.g., buckling / kink margin tests)
VF-004	fatigue life (run-out) testing. Acceptance criteria: No failure observed under specified loading conditions.	fatigue life (run-out) testing (e.g., fatigue life (run-out) testing)	fatigue life (run-out) testing (e.g., fatigue life (run-out) testing)	fatigue life (run-out) testing (e.g., fatigue life (run-out) testing)
VF-005	buckling/kink margin test across the duty cycle	buckling/kink margin test across the duty cycle (e.g., buckling/kink margin test across the duty cycle)	buckling/kink margin test across the duty cycle (e.g., buckling/kink margin test across the duty cycle)	buckling/kink margin test across the duty cycle (e.g., buckling/kink margin test across the duty cycle)
VF-006	fatigue run-out at worst-case loading	fatigue run-out at worst-case loading (e.g., fatigue run-out at worst-case loading)	fatigue run-out at worst-case loading (e.g., fatigue run-out at worst-case loading)	fatigue run-out at worst-case loading (e.g., fatigue run-out at worst-case loading)
VF-007	buckling / kink margin tests across duty cycle	buckling / kink margin tests (e.g., buckling / kink margin tests)	buckling / kink margin tests (e.g., buckling / kink margin tests)	buckling / kink margin tests (e.g., buckling / kink margin tests)
VF-008	overload margin demonstration to worst-case design	overload margin demonstration to worst-case design (e.g., overload margin demonstration to worst-case design)	overload margin demonstration to worst-case design (e.g., overload margin demonstration to worst-case design)	overload margin demonstration to worst-case design (e.g., overload margin demonstration to worst-case design)

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification passes)	criteria to be fixed at NOT_DONE	NOT PERFORMED Definition; validation is

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	nitinol / polymer composite membrane	minimally invasive device practice	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
nitinol / polymer composite membrane	nitinol / polymer composite membrane	1	Material	nitinol / polymer composite membrane	NOT ESTABLISHED	NOT ESTABLISHED	Reliability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- laser-cut hypotube / braided composite shaft: domain registry manufacturing route - candidate only; no process qualification exists
- passivation and surface finish control: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter member slenderness ratio [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter yield/ultimate strength [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter applied cyclic load amplitude [P_cr N]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter required run-out cycles [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter articulation radius [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter friction coefficient at guides [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'acceptance criterion for verification VF-001', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-001'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}

- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}
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- {'unknown': 'acceptance criterion for verification VF-006', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-006'}
- {'unknown': 'acceptance criterion for verification VF-007', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-007'}
- {'unknown': 'acceptance criterion for verification VF-008', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-008'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	maintaining the proposed envelope	constraint-compliance	NOT ESTABLISHED	constraint-compliance	NOT ESTABLISHED
WP-02	constraint-compliance envelope article	per method	Fatigue testing of topology	NOT ESTABLISHED	VF-001 verification	NOT ESTABLISHED
WP-03	article for VF-001	per method	run killer experiment	NOT ESTABLISHED	VF-002 verification	NOT ESTABLISHED
WP-04	article for VF-002	per method	buckling / kink margin	NOT ESTABLISHED	VF-003 verification	NOT ESTABLISHED
WP-05	article for VF-003	per method	fatigue life (run-out) test	NOT ESTABLISHED	VF-004 verification	NOT ESTABLISHED
WP-06	article for VF-004	per method	buckling/kink margin	NOT ESTABLISHED	VF-005 verification	NOT ESTABLISHED
WP-07	article for VF-005	per method	fatigue run-out at worst	NOT ESTABLISHED	VF-006 verification	NOT ESTABLISHED
WP-08	article for VF-006	per method	buckling / kink margin	NOT ESTABLISHED	VF-007 verification	NOT ESTABLISHED
WP-09	article for VF-007	per method	overload margin	NOT ESTABLISHED	VF-008 verification	NOT ESTABLISHED
WP-10	article for VF-008	per method				

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 10-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.